20/08/2025

Code-B

Aakash
Medical | IIT-JEE | Foundations

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MM : 720 Fortnightly Test for NEET-2026_RM(P2)_FT-03B **Time : 180 Min.**

Topics Covered:
Physics: Work, Energy & Power, System of Particles & Rotational Motion
Chemistry: Chemical Bonding and Molecular Structure
Botany: Biological Classification
Zoology: Breathing & Exchange of Gases-II: (Upto disorders), Body Fluids & Circulation

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

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PHYSICS

1. A : Internal force can change the KE of a system.
R : Internal force cannot change the momentum of system.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

2. A body constrained to move along the *y*-axis of a coordinate system is subjected to constant force (in *N*) given by $\vec{F} = -\hat{i} + 2\hat{j} + 3\hat{k}$, where $\hat{i}$, $\hat{j}$ and $\hat{k}$ are unit vectors along the *x*-axis, *y*-axis and *z*-axis respectively. The work done by this force in displacing the body by 4 m will be

(1) 15 J

(2) 8 J

(3) 10 J

(4) 20 J

3. Work done by friction is

(1) Always zero

(2) Always positive

(3) May be positive, negative or zero

(4) Always negative

4. A body of mass 6 kg is lowered down with the help of a string with downward acceleration $\frac{g}{4}$. The work done by the string on the body in first 3 s after the start of the motion is (g = 10 m/s^2)

(1) –675 J

(2) –506.25 J

(3) –2025 J

(4) Zero

5. A projectile of mass *m* is projected at an angle θ with horizontal with a speed *u*. The work done by gravity from point of projection to maximum height is

(1) $\frac{mu^2 \sin^2\theta}{2}$

(2) $-\frac{mu^2 \sin^2\theta}{2}$

(3) $2mu^2 \sin^2\theta$

(4) $-2mu^2 \sin^2\theta$

6. A small block of mass *m* is kept on a rough inclined surface of inclination θ fixed in an elevator. The elevator goes up with a uniform velocity *v* and the block does not slide on the wedge. The work done by the normal contact force between the block and the wedge in time *t* is

(1) *mgvt* cosθ

(2) $mgvt\cos^2\theta$

(3) $mgvt\sin^2\theta$

(4) *mgvt*

7. The work done by all the forces (external and internal) on the system equals the change in

(1) Total energy

(2) Kinetic energy

(3) Potential energy

(4) None of these

8. Under the action of a force, a mass of 4 kg moves rectilinearly such that its position (*x*) varies with time (*t*) as $x = \frac{1}{4}t^3$. The work done by the force between $t = 1$ s and $t =$ 3 s will be

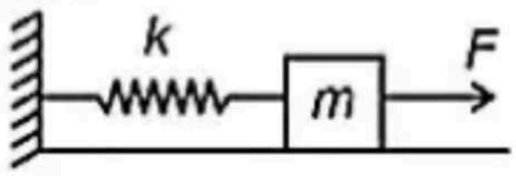
(1) 85 J

(2) 90 J

(3) 63 J

(4) 115 J

9. A block of mass *m* is fixed to a spring of spring constant *k* which is in its natural length. If the surface is smooth and a constant force *F* is applied on the block as shown in figure, then



(1) Maximum elongation in spring is $\frac{F}{k}$

(2) Maximum elongation in spring is $\frac{2F}{k}$

(3) Maximum elongation in spring is $\frac{3F}{k}$

(4) Maximum elongation in spring is $\frac{F}{3k}$

10. A force $F = (2 + 2x)$ N acts on a particle in positive *x*-direction. Increase in kinetic energy of block when it moves from $x = 0$ to $x = 1$ m is {*x* is measured in meter}

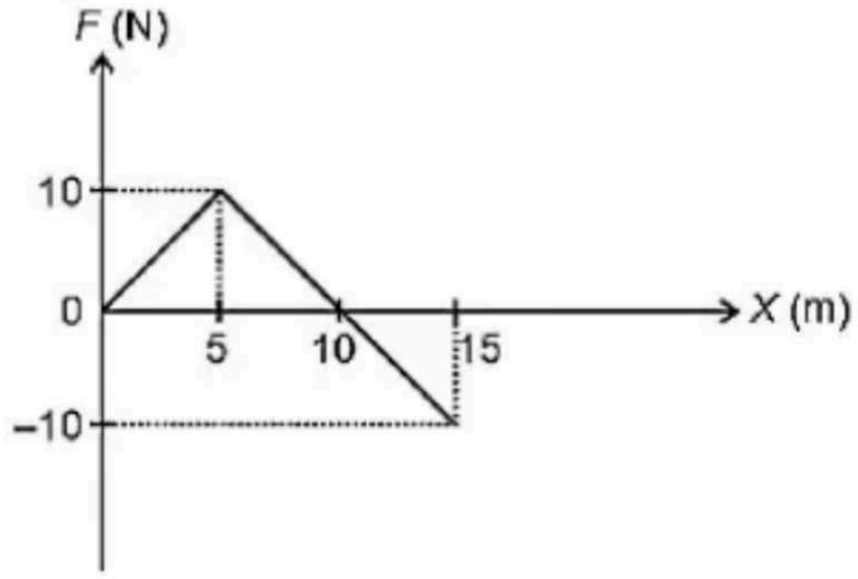
(1) 1 J

(2) 3 J

(3) 4 J

(4) 6 J

11. The force (*F*) versus displacement (*X*) graph of a moving object is given below. If the object was initially at rest at $x = 0$, the total work done by the force on the object from $x = 0$ to $x = 15$ m is



(1) 50 J

(2) 25 J

(3) 75 J

(4) −25 J

12. In a conservative field, the potential energy *U* as a function of position *x* is given as $U = x^2 + x + 3$, then the corresponding conservative force is given by

(1) $2x + 1$

(2) $-2x + 1$

(3) $2x + 3$

(4) $-2x - 1$

13. A uniform chain of mass *M* and length *L* is placed on table such that $\frac{1}{4}$th part of its length is hanging. The amount of work done in bringing this hanging part onto the table is

(1) $\frac{MgL}{32}$

(2) $\frac{MgL}{64}$

(3) $\frac{MgL}{8}$

(4) $\frac{MgL}{16}$

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14. A : Positive work done by a conservative force is equal to decrease in potential energy of a system.
R : Potential energy is not defined for non-conservative-field.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

15. A stone of mass 1 kg is tied to a string and it is whirled in a vertical circle of radius 2 m. If tension at highest point is 15 N, then the velocity at lowest point is nearly

(1) $\sqrt{130}$ ms^{-1}

(2) $\sqrt{150}$ ms^{-1}

(3) $\sqrt{170}$ ms^{-1}

(4) $\sqrt{190}$ ms^{-1}

16. A : In elastic collision, KE of the bodies remain unchanged.
R : During collision, momentum of colliding bodies are zero.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

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17. Power supplied to a particle of mass 1 kg varies with time as $P = \frac{3t^2}{2}$ watt, t is in seconds and initially particle was at rest. Velocity of the particle at t = 3 s is

(1) 2 ms^{-1}

(2) 4 ms^{-1}

(3) $3\sqrt{3}$ ms^{-1}

(4) $\sqrt{6}$ ms^{-1}

18. A motor pump, pumps 300 kg of water through height of 20 m per minute. If efficiency of motor is 80% then input power of motor pump is (g = 10 ms^{-2})

(1) 1.25 kW

(2) 1 kW

(3) 2.50 kW

(4) 2 kW

19. A force $\left(4\hat{i}+\hat{j}-3\hat{k}\right)$ N acting on a body maintains its velocity at $\left(2\hat{i}+2\hat{j}+2\hat{k}\right)$ m/s. The instantaneous power delivered by the force is

(1) 4 W

(2) 8 W

(3) 16 W

(4) 10 W

20. CGS unit of work is

(1) Slug

(2) Dyne

(3) Erg

(4) Joule

21. From the state of rest, a car accelerates while engine supplies constant power P. The velocity of car varies with time as

(1) $v \propto t$

(2) $v \propto t^{\frac{1}{2}}$

(3) $v \propto t^2$

(4) $v \propto t^{\frac{3}{2}}$

22. A ball is dropped from height 20 m above the ground. Height attained by ball after first collision with ground will be (Coefficient of restitution, e = 0.5)

(1) 4 m

(2) 5 m

(3) 8 m

(4) 12 m

23. A ball hits the floor with speed v making an angle of incidence θ with the normal. If the coefficient of restitution is e, then angle of reflection of the ball is

(1) $\sin^{-1}\left(\frac{\sin\theta}{e}\right)$

(2) $\tan^{-1}\left(\frac{\tan\theta}{e}\right)$

(3) $\cot^{-1}\left(\frac{\cot\theta}{e}\right)$

(4) $\cot^{-1}\left(\frac{\sin\theta}{e}\right)$

24. Two bodies *A* and *B* of the same mass undergo completely inelastic one dimensional collision. The body *A* was moving with kinetic energy K_1 while body *B* was at rest before collision. The kinetic energy of the system after collision is K_2. The ratio $K_2 : K_1$ is

(1) 1 : 2

(2) 1 : 4

(3) 2 : 3

(4) 4 : 3

25. Two bodies of masses 4 kg and 6 kg are lying in *XY* plane at (1, 2) and (3, 4) respectively. The coordinates of centre of mass of the two body system are

(1) $\left(\frac{11}{9}, \frac{16}{5}\right)$

(2) $\left(\frac{11}{5}, \frac{16}{5}\right)$

(3) $\left(\frac{7}{9}, \frac{16}{5}\right)$

(4) $\left(\frac{16}{5}, \frac{7}{9}\right)$

26. A bomb of mass 12 kg is projected upward with velocity of 50 m/s. It explodes after 2 s into two fragments, 3 kg and 9 kg. If 9 kg mass comes to rest, then velocity of other fragment is

(1) 60 m/s

(2) 120 m/s

(3) 100 m/s

(4) 80 m/s

27. A force $\vec{F} = (2\hat{i} + 3\hat{j} + 4\hat{k})$ N is acting at point $(2\hat{i} + 2\hat{j} + 2\hat{k})$ m . The magnitude of its torque about origin will be

(1) $4\sqrt{6}$ N m

(2) $2\sqrt{6}$ N m

(3) $6\sqrt{2}$ N m

(4) $3\sqrt{2}$ N m

28. A man of mass 60 kg is sitting at one end of the boat of mass 60 kg and length 3 m. The boat is at rest in the still water. If the man starts walking on the boat and reaches the other end then the displacement of boat w.r.t. ground is

(1) 1 m

(2) 1.5 m

(3) 3 m

(4) 2.5 m

29. **Statement A :** Algebraic sum of moments of masses about the centre of mass is zero.
Statement B : For small bodies, on earth's surface, centre of mass coincides with centre of gravity.

(1) Only A is correct

(2) Only B is correct

(3) Both A and B are correct

(4) Neither A nor B are correct

30. From a uniform disc of radius *R*, a smaller disc of radius $\frac{R}{2}$ is removed. The centre of the hole formed is at $\frac{R}{2}$ from the centre of the original disc. The shift in position of centre of mass of the disc after removing smaller disc will be

(1) $\frac{R}{3}$

(2) $\frac{R}{2}$

(3) $\frac{R}{4}$

(4) $\frac{R}{6}$

31. Two bodies of different masses of 2 kg and 4 kg are moving with speed 20 m/s and 10 m/s towards each other due to mutual gravitational attraction. The speed of their centre of mass is

(1) 5 m/s

(2) 6 m/s

(3) 16

(4) 8 m/s

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32. The linear density of a thin rod of length 1 m varies as $\lambda = 3x^2 + 2$, where *x* is the distance from its one end. The distance of centre of mass from this end is

(1) $\frac{1}{2}m$

(2) $\frac{3}{4}m$

(3) $\frac{5}{12}$ m

(4) $\frac{7}{12}$ m

33. Match **list I** with **list II** and tick the correct option (where MOI implies moment of inertia).

	List I		List II
a.	MOI of a thin rod (M, L) about its COM axis which is perpendicular to its length	(i)	$\frac{MR^2}{4}$
b.	MOI of a uniform ring (M, R) about its axis	(ii)	$\frac{ML^2}{3}$
c.	MOI of a solid sphere (M, R) about its diametrical axis	(iii)	MR^2
d.	MOI of a uniform disc (M, R) about its diametrical axis	(iv)	$\frac{2}{5}MR^2$
		(v)	$\frac{ML^2}{12}$

(1) a(v), b(iii), c(iv), d(i)

(2) a(ii), b(iii), c(i), d(iv)

(3) a(v), b(iii), c(i), d(iv)

(4) a(ii), b(i), c(iii), d(iv)

34. A constant torque of 100 N m is acting on a rigid body of moment of inertia 20 kg m^2 about an axis passing through the centre. Its angular velocity after 4 s starting from rest will be

(1) 10 rad/s

(2) 20 rad/s

(3) 40 rad/s

(4) Zero

35. If angular velocity of disc depends on angle rotated θ as $\omega = \theta^2 + 2\theta + 1$ (where ω is in rad/s and θ is in rad), then its angular acceleration at θ = 1 rad is

(1) 12 rad/s^2

(2) 16 rad/s^2

(3) 10 rad/s^2

(4) 4 rad/s^2

36. A cubical block of mass M and edge length l slides down a rough inclined plane of inclination θ with a uniform velocity. The torque of the normal force on the block about the centre of cube has a magnitude

(1) Zero

(2) $\frac{1}{2}mgl\cos\theta$

(3) $\frac{1}{2}mgl\sin\theta$

(4) $mgl\cos\theta$

37. A particle of mass 2 kg is moving with uniform speed $4\sqrt{2}$ m/s in x–y plane along the line $y = x + 3$. The magnitude of its angular momentum about the origin is

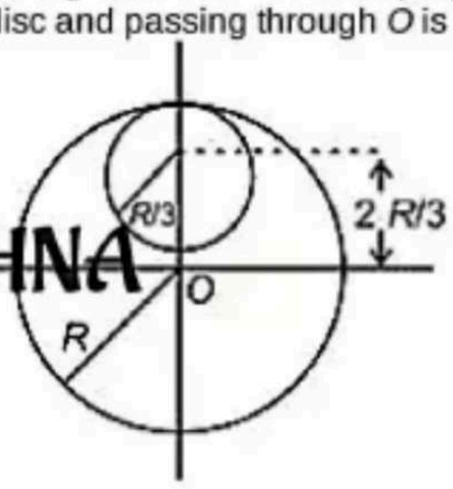
(1) Zero

(2) 24 kg m^2s^{-1}

(3) 48 kg m^2s^{-1}

(4) 12 kg m^2s^{-1}

38. From a circular disc of Radius R and mass $9M$, a small disc of radius $\frac{R}{3}$ is removed. The moment of inertia of the remaining disc about an axis perpendicular to the plane of the disc and passing through O is



(1) $4MR^2$

(2) $\frac{40}{9}MR^2$

(3) $10MR^2$

(4) $\frac{37}{9}MR^2$

39. The radius of gyration of a uniform circular disc of radius a, rotating about an axis perpendicular to the plane and passing through its centre of mass, is

(1) $\frac{a}{\sqrt{3}}$

(2) $\frac{a}{2}$

(3) $\frac{a}{\sqrt{2}}$

(4) $\frac{a}{3}$

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40. A rod of mass *m* and length *l* is hinged about its one of the ends. The rod is released from horizontal position, the angular speed of the rod when the rod becomes vertical is:

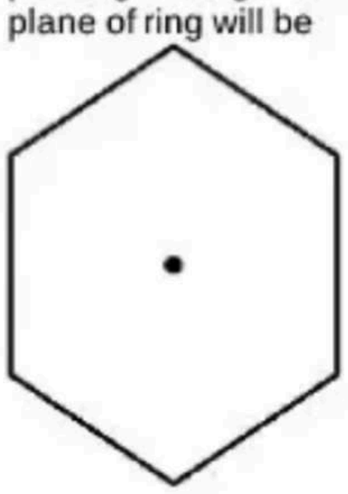
(1) $\omega = \sqrt{\frac{3g}{L}}$

(2) $\omega = \sqrt{\frac{5g}{L}}$

(3) $\omega = \sqrt{\frac{7g}{L}}$

(4) $\omega = 2\sqrt{\frac{2g}{L}}$

41. Six identical thin rods are joined to form a closed structure in the form of a Benzene ring. If each rod has mass *M* and length *L* then moment of inertia of this ring about an axis passing through centre of ring and perpendicular to the plane of ring will be



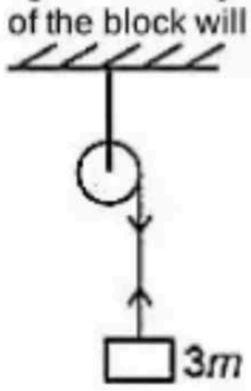
(1) $\frac{33}{12}ML^2$

(2) $11ML^2$

(3) $\frac{5}{6}ML^2$

(4) $5ML^2$

42. A block of mass $3m$ is connected with a string which is wound on pulley of mass *m* and radius *R* as shown in the figure. If the system is released from rest. Then acceleration of the block will be (consider pulley as uniform disc)



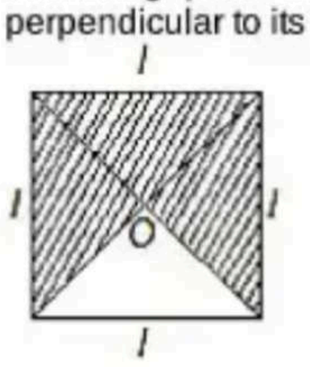
(1) g

(2) $\frac{5g}{3}$

(3) $\frac{6g}{7}$

(4) $\frac{g}{3}$

43. An isosceles triangular piece is cut from a square plate of side *l*. The piece is one-fourth of the square and mass of the remaining plate is *m*. The moment of inertia of the remaining plate about an axis passing through *O* and perpendicular to its plane is



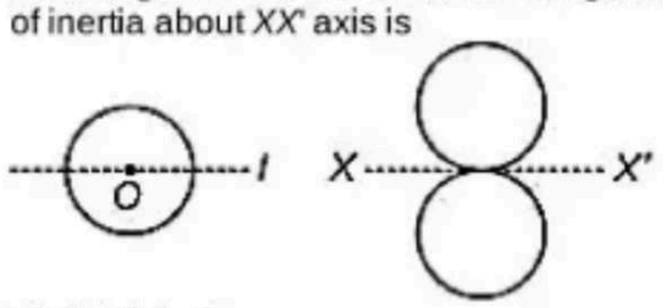
(1) $\frac{ml^2}{6}$

(2) $\frac{ml^2}{12}$

(3) $\frac{ml^2}{24}$

(4) $\frac{ml^2}{3}$

44. Moment of inertia of a ring about its diameteric axis is *I*. Two such rings are welded as shown in figure. The new moment of inertia about *XX'* axis is



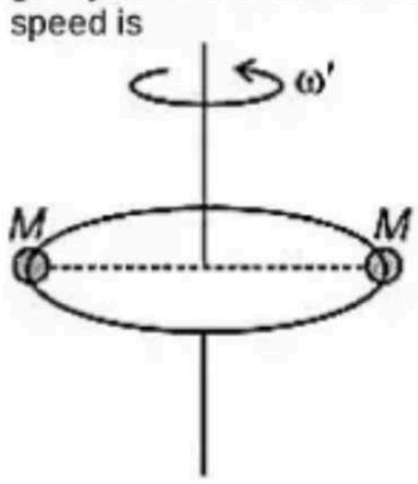
(1) I

(2) $3I$

(3) $2I$

(4) $\frac{I}{2}$

45. A disc of mass *M* and radius *R* rotates with an angular speed ω. If two point masses each of mass *M* are placed gently at a distance *R* from the centre. Then new angular speed is



(1) 3ω

(2) $\frac{\omega}{5}$

(3) $\frac{\omega}{3}$

(4) 5ω

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CHEMISTRY

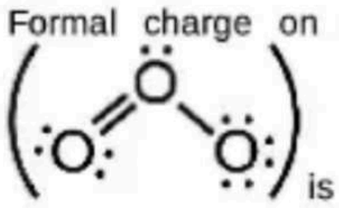
46. Given below are the two statements
Statement I : O_3 molecule is a resonance hybrid of three canonical structures.
Statement II : Resonance stabilizes the molecule as the energy of the resonance hybrid is less than the energy of any single canonical structure.
In light of above statements, choose the correct answer.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

47. Formal charge on middle oxygen of ozone molecule  is

(1) 0

(2) +1

(3) –1

(4) –2

48. Which of the following is an odd electron species?

(1) CH_4

(2) SF_6

(3) NO

(4) BF_3

49. The correct order of decreasing bond length of C–C, C–H, C–O and O–H is

(1) C–C > C–O > O–H > C–H

(2) C–O > C–C > C–H > O–H

(3) C–C > C–O > C–H > O–H

(4) C–O > C–C > O–H > C–H

50. The correct order of ionic character is

(1) $KCl < CaCl_2 < ZnCl_2$

(2) $ZnCl_2 < CaCl_2 < KCl$

(3) $KCl < ZnCl_2 < CaCl_2$

(4) $ZnCl_2 < KCl < CaCl_2$

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51. Given below are two statements one is labelled as assertion (A) and other is labelled as Reason (R)
Assertion (A): Dipole moment of NF_3 is greater than that of NH_3.
Reason (R): F is more electronegative than H.
In the light of the above statements, choose the correct answer from the options given below

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is false but (R) is true

(4) (A) is true but (R) is false

52. Match species given in List-I with their bond angles in List-II

	List I		List-II
a.	CH_4	(i)	107°
b.	NH_3	(ii)	109.5°
c.	BF_3	(iii)	104.5°
d.	H_2O	(iv)	120°

Choose the correct option

(1) a(ii), b(i), c(iv), d(iii)

(2) a(i), b(ii), c(iii), d(iv)

(3) a(ii), b(iv), c(i), d(iii)

(4) a(iv), b(ii), c(iii), d(i)

53. If the observed dipole moment of A–B molecule is 1 D and A – B bond length is 100 pm then percentage ionic character in A–B molecule will be

(1) 25.75%

(2) 16.65%

(3) 20.81%

(4) 14.29%

54. Consider the following molecules
PF_5, BF_3, IF_7, NO_2, H_3PO_4, SF_4, SO_3
The total number of molecules which have expanded octet is

(1) 5

(2) 3

(3) 4

(4) 6

55. Consider the following two statements
Statement I : Formal charges help in the selection of the lowest energy structure from a number of possible Lewis structures for a given species.
Statement II : Generally the, lowest energy structure is the one with the maximum formal charges on atoms.
In the light of above statements, choose the correct option.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

56. Correct order of dipole moment of the given molecules is

(1) $CO_2 > H_2S > H_2O$

(2) $H_2O > CO_2 > H_2S$

(3) $CO_2 > H_2O > H_2S$

(4) $H_2O > H_2S > CO_2$

57. Bond order of N_2^- and O_2^- respectively are

(1) 2.5 and 2

(2) 3 and 1.5

(3) 2.5 and 1.5

(4) 1.5 and 2.5

58. The bond order of N – O bond and P – O bond in NO_3^- and PO_4^{3-} ion respectively are

(1) 1.33 and 1.25

(2) 1.5 and 1.25

(3) 1.25 and 1.5

(4) 1.5 and 1.33

59. Lowest lattice energy will be possessed by

(1) NaF

(2) MgF_2

(3) AlF_3

(4) LiF

60. Number of carbon atoms having *sp* hybridization in the given molecule is
$CH_3 - C \equiv C - CH = CH_2$

(1) 2

(2) 3

(3) 4

(4) 5

61. Electrovalency of calcium in $CaCl_2$ is

(1) +1

(2) +2

(3) Zero

(4) –2

62. Match **List-I** with **List-II**

List-I		**List-II**
A. XeO_3	(I)	sp^3d; linear
B. XeF_2	(II)	sp^3; pyramidal
C. $XeOF_4$	(III)	sp^3d^3; distorted octahedral
D. XeF_6	(IV)	sp^3d^2; square pyramidal

Choose the **correct** answer from the options given below :

(1) A-II, B-I, C-III, D-IV

(2) A-IV, B-II, C-III, D-I

(3) A-IV, B-II, C-I, D-III

(4) A-II, B-I, C-IV, D-III

63. Number of lone pair(s) of electrons on central atom and the shape of ClF_3 molecule respectively, are

(1) 0, triangular planar

(2) 1, pyramidal

(3) 2, bent T-shape

(4) 2, T-shape

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64. Identify the **incorrect** statement for PCl_5 from the following.

(1) Orbitals of phosphorus are assumed to undergo sp^3d hybridization

(2) PCl_5 has two axial bonds stronger than three equational bonds

(3) The three equatorial bonds of PCl_5 lie in a plane

(4) The geometry of PCl_5 is trigonal bipyramidal

65. Given below are two statements
Statement I : Bond enthalpy of N_2 molecule is higher than O_2 molecule.
Statement II : Bond enthalpy of HCl molecule is higher than H_2 molecule.
In the light of above statements, choose the correct answer from the options given below

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

66. Given below are two statements
Statement (I): The number of hybrid orbitals is equal to number of atomic orbitals those get hybridised.
Statement (II): The hybridised orbitals differ in shape and enegy.
In the light of above statements, choose the most appropriate answer from the options given below.

(1) Both statement (I) and statement (II) are correct.

(2) Both statement (I) and statement (II) are incorrect.

(3) Statement (I) is correct but statement (II) is incorrect.

(4) Statement (I) is incorrect but statement (II) is correct.

67. The pair of hybridised orbitals involved in the formation of $C_2 - C_3$ bond in $CH_2 = CH - CH_2 - C \equiv CH$ is

(1) $sp - sp^3$

(2) $sp^2 - sp^3$

(3) $sp^3 - sp^3$

(4) $sp - sp$

68. Number of $p\pi - p\pi$ bonds present in SO_3 molecule is

(1) Zero

(2) 1

(3) 2

(4) 3

69. Number of 90° bond angles in SF_6 molecule is

(1) 4

(2) 8

(3) 10

(4) 12

70. Correct decreasing order of the repulsive interaction of electron pair is

(1) Lone pair–lone pair > Lone pair –bond pair > bond pair–bond pair

(2) Lone pair–bond pair > Lone pair –lone pair > bond pair–bond pair

(3) Lone pair–bond pair > bond pair –bond pair > lone pair–lone pair

(4) Bond pair–bond pair > Lone pair –bond pair > Lone pair–lone pair

71. Percentage '*p*' character in sp^3d hybrid orbital is

(1) 20%

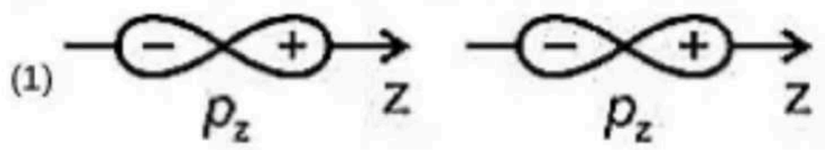
(2) 60%

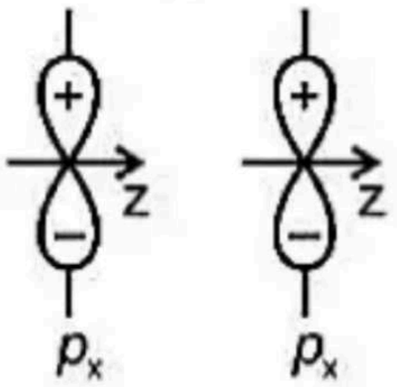
(3) 33%

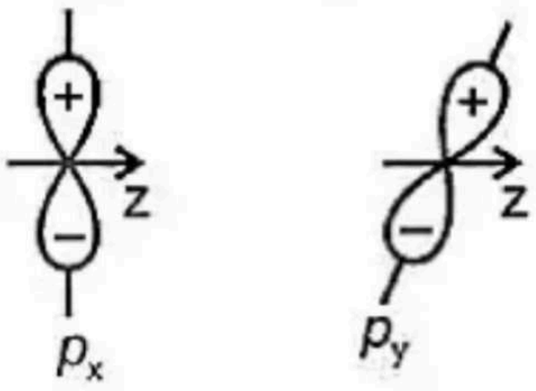
(4) 40%

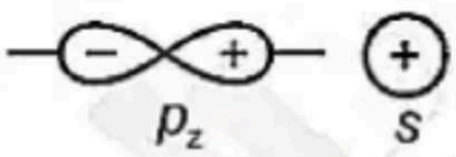
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72. Which of the following results into zero overlap?

(1) 

(2) 

(3) 

(4) 

73. Correct order of polarising power of the given ions is

(1) $Be^{2+} < K^+ < Ca^{2+} < Mg^{2+}$

(2) $Mg^{2+} < Be^{2+} < K^+ < Ca^{2+}$

(3) $Ca^{2+} < Mg^{2+} < Be^{2+} < K^+$

(4) $K^+ < Ca^{2+} < Mg^{2+} < Be^{2+}$

74. Consider the following statements
(a) O_2 molecule is paramagnetic in nature.
(b) N_2^+ has bond order 2.5.
(c) Li_2^{2+} does not exist
The correct statements are

(1) (a) and (b) only

(2) (b) and (c) only

(3) (a) and (c) only

(4) (a), (b) and (c)

75. Given below are two statements
Statement I : According to molecular orbital theory, both C_2 and B_2 species have only π bonds
Statement II : Both $\pi 2p_x$ and $\pi 2p_y$ molecular orbitals of B_2 and C_2 have 2 electrons each
In the light of above statements, choose the correct answer from the options given below

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

76. In which of the following processes, the stability increases and the magnetic behaviour changes?

(1) $Li_2 \rightarrow Li_2^+$

(2) $O_2 \rightarrow O_2^+$

(3) $N_2^+ \rightarrow N_2$

(4) $H_2 \rightarrow H_2^+$

77. Correct order of C–O bond length in given species is

(1) $CO_3^{2-} < CO_2 < CO$

(2) $CO < CO_3^{2-} < CO_2$

(3) $CO_2 < CO < CO_3^{2-}$

(4) $CO < CO_2 < CO_3^{2-}$

78. The correct order of bond strength among the species is

(1) $O_2^+ > O_2 > O_2^- > O_2^{2-}$

(2) $O_2^{2-} > O_2^- > O_2 > O_2^+$

(3) $O_2 > O_2^+ > O_2^- > O_2^{2-}$

(4) $O_2 > O_2^- > O_2^{2-} > O_2^+$

79. During formation of O_2 from O_2^-, electron is released from the orbital

(1) $\sigma 2p_z$

(2) $\sigma^* 2p_z$

(3) $\pi 2p_x$

(4) $\pi^* 2p_x$

80. Which of the following species is see-saw in shape?

(1) SF_2Cl_2

(2) $[ICl_4]^-$

(3) PCl_5

(4) BrF_5

81. Which among the following is not a planar molecule?

(1) BCl_3

(2) C_2H_4

(3) SF_4

(4) H_2O

82. Which of the following pairs of compounds is isostructural?

(1) NH_3, BF_3

(2) $BeCl_2$, ICl_3

(3) BCl_3, ClF_3

(4) I_3^-, XeF_2

83. Overlapping of which pair of orbitals form π-bond (If x-axis is intermolecular axis)?

(1) $p_x + p_y$

(2) $p_x + p_x$

(3) $p_y + p_y$

(4) $s + p_x$

84. Number of anti-bonding electrons in a F_2 molecule is

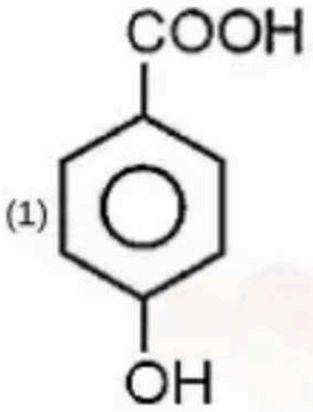
(1) 8

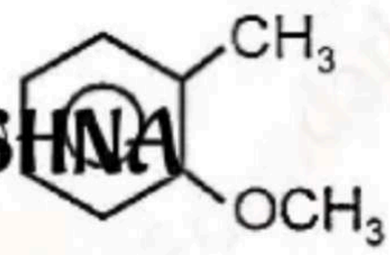
(2) 6

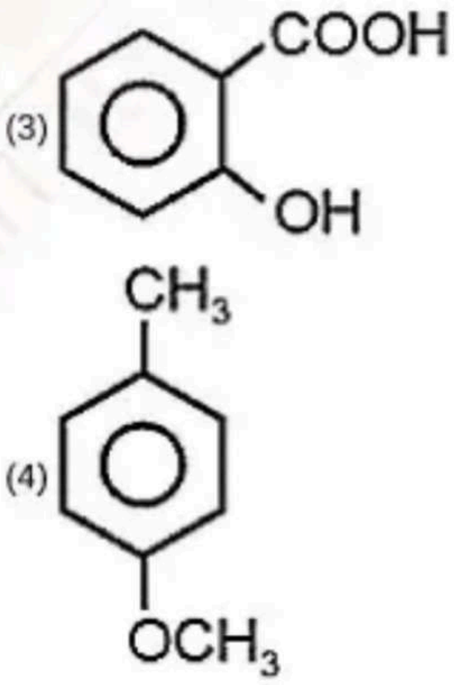
(3) 2

(4) 4

85. Intramolecular hydrogen bond is present in

(1) 

(2) 

(3) 

(4)

86. Which of the following molecular orbitals contain zero nodal plane?

(1) $\sigma^* 1s$

(2) $\sigma 2s$

(3) $\pi 2p_x$

(4) $\pi^* 2p_x$

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87. Maximum number of atoms lie in a plane for CCl_4 and SF_6 molecules respectively are

(1) 4 and 4

(2) 3 and 5

(3) 3 and 4

(4) 5 and 7

88. Given below are the two statements
Statement I: The magnitude of H-bonding depends on the physical state of the compound.
Statement II: The magnitude of H-bonding is maximum in gaseous state and minimum in solid state.
In the light of above statements, choose the correct answer.

(1) Statement I is correct but Statement II is incorrect

(2) Both Statement I and Statement II are correct

(3) Both Statement I and Statement II are incorrect

(4) Statement I is incorrect but Statement II is correct

89. Molecule that does not exist is

(1) C_2

(2) B_2

(3) He_2

(4) F_2

90. Consider the following statements
Statement I : Ethanol forms intermolecular hydrogen bond with each other.
Statement II : o-nitrophenol forms intramolecular hydrogen bond.
In the light of above statements choose the correct option.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

BOTANY

91. Select the **correct** set of statements for the virus first crystallised by Stanley.
(a) It has either single or double stranded DNA as its genetic material.
(b) Subunits of their protein coat are arranged helically
(c) It is an obligate plant parasite.
(d) First discovered by M.W. Beijerinck.

(1) Only (b) and (c)

(2) Only (a) and (b)

(3) All except (d)

(4) All (a), (b), (c) and (d)

92. Puff balls and toadstools belong to class

(1) Phycomycetes

(2) Ascomycetes

(3) Basidiomycetes

(4) Deuteromycetes

93. Who demonstrated that the extract of infected plants of tobacco could cause infection in healthy plants and called this fluid as '*Contagium vivum fluidum*'?

(1) W. M. Stanley

(2) D. J. Ivanowsky

(3) M. W. Beijerinck

(4) Pasteur

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94. Match the column I with column II and select the **correct** option.

Column I	Column II
a. Methanogens	(i) Photoautotrophic
b. Halophiles	(ii) Guts of ruminants
c. Thermoacidophiles	(iii) Heterotrophic
d. Cyanobacteria	(iv) Hot water springs

(1) a(ii), b(iii), c(i), d(iv)

(2) a(ii), b(iii), c(iv), d(i)

(3) a(iii), b(ii), c(i), d(iv)

(4) a(iii), b(ii), c(iv), d(i)

95. Select the **correct** match.

(1) Amoeboid protozoans – *Trypanosoma*

(2) *Sporozoans* – *Plasmodium*

(3) Ciliated protozoans – *Entamoeba*

(4) Flagellated protozoans – *Paramoecium*

(1) (1)

(2) (2)

(3) (3)

(4) (4)

96. Which of the following does **not** belong to kingdom Protista?

(1) *Euglena*

(2) Dinoflagellates

(3) *Amoeba*

(4) *Spirulina*

97. Select **incorrect** statement w.r.t. *Puccinia*.

(1) It is rust causing fungi

(2) Sex organs are present

(3) Mycelium is branched and septate

(4) Vegetative reproduction takes place by fragmentation

98. State the given statements as **True (T)** or **False (F)** and select the **correct** option.
a. Cr-Jacob disease (CJD) in humans is caused by prions.
b. Infectious agent that causes potato spindle tuber disease lacks the protein coat which is found in viruses.
c. The algal component of lichens absorbs mineral nutrient and water for its partner.

(1) a-T, b-T, c-T

(2) a-T, b-T, c-F

(3) a-F, b-F, c-T

(4) a-T, b-F, c-F

99. Which of the given fungus is used extensively in biochemical and genetic work?

(1) *Neurospora*

(2) *Alternaria*

(3) *Puccinia*

(4) *Penicillium*

100. Read the following statements for the given figure and select the **correct** option.



a. It is a member of basidiomycetes.
b. Its mycelium is branched and septate.
c. Sex organ are present.
d. It produces endogenous sexual spores.

(1) a and c

(2) a and b

(3) c and d

(4) b and d

101. Select the **incorrectly** matched pair.

(1) *Nostoc* – Has heterocyst

(2) *Euglena* – Has lipid-rich layer pellicle

(3) *Bacillus* – Is rod-shaped bacteria

(4) Dinoflagellates – Form red tides in sea

102. Which of the following is **incorrect** w.r.t heterotrophic bacteria?

(1) Majority of them are important decomposers

(2) Helpful in production of antibiotics

(3) Fix the nitrogen in legumes

(4) Includes the blue-green algae (BGA)

103. Bacteria do not have

(1) 70 S ribosomes

(2) Nucleoid

(3) Membrane bound cell organelles

(4) Flagella

104. Bacterial viruses usually contain

(1) Double stranded RNA

(2) Single stranded RNA

(3) Double stranded DNA

(4) Single stranded DNA

105. In five kingdom classification, *Chlamydomonas* is placed in kingdom

(1) Monera

(2) Plantae

(3) Animalia

(4) Protista

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106. Read the following statements and select the **correct** option.
Statement A: All protozoans are heterotrophs.
Statement B: Protozoans are believed to be primitive relatives of animals.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements are incorrect

(4) Both statements are correct

107. Fruiting body of *Aspergillus* is called

(1) Conidium

(2) Basidiocarp

(3) Ascocarp

(4) Sporangium

108. All of the given organisms represent multicellular fungi, **except**

(1) *Penicillium*

(2) Yeast

(3) *Agaricus*

(4) *Claviceps*

109. Bovine spongiform encephalopathy is caused by

(1) An agent which is abnormally folded protein

(2) An RNA virus

(3) A bacteriophage

(4) An agent that contains RNA of low molecular weight

110. Nucleic acid is absent in a

(1) Virus

(2) Viroid

(3) Prion

(4) Bacterium

111. Cell wall is impregnated with silica in

(1) Cellular slime moulds

(2) Diatoms

(3) Methanogens

(4) Dinoflagellates

112. Which of the following pairs of fungi are edible and considered delicacies?

(1) Morels and *Alternaria*

(2) *Albugo* and *Claviceps*

(3) Morels and Truffles

(4) *Yeast* and *Agaricus*

113. Read the following statements w.r.t. sac fungi

A. Conidia are produced exogenously
B. Sex organs are absent
C. Ascus is the site of karyogamy and meiosis
D. Fruiting bodies are called ascocarps
E. Ascospores are produced exogenously
F. Some members are coprophilous

Mark the option representing **incorrect** statements

(1) B & E

(2) C, E & F

(3) B, E & F

(4) D, F & C

114. Which of the given character is shown by all members of protozoa?

(1) Unicellular

(2) Autotrophic nutrition

(3) Spore formation

(4) Silica shells

115. Cell walls form two thin overlapping shells, which fit together as in a soap box, in

(1) Euglenoids

(2) Diatoms

(3) Yeast

(4) *Trypanosoma*

116. Select the **wrong** statement about chemoautotrophic bacteria.

(1) They do not have photosynthetic pigments

(2) They use light as energy source for synthesizing food

(3) They oxidise certain inorganic substances

(4) They play a great role in recycling of nutrients like nitrogen

117. Which of the given is a chief producer in ocean?

(1) Euglenoids

(2) Dinoflagellates

(3) Protozoans

(4) Diatoms

118. Given below are two statements.

Assertion (A): Members of Ascomycetes and Basidiomycetes have an intervening dikaryotic stage.
Reason (R): In these fungi plasmogamy is not immediately followed by karyogamy.
In the light of the above statements, choose the **correct** answer from the options given below.

(1) (A) is true but (R) is false

(2) Both (A) and (R) are true and (R) is the correct explanation of (A)

(3) (A) is false but (R) is true

(4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

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119. Read the following statements and find out the **correct** one w.r.t. slime moulds.

(1) Their body moves along decaying twigs and leaves engulfing inorganic materials

(2) During favourable conditions they form fruiting bodies

(3) Their spores possess true walls

(4) During unfavourable conditions they form an aggregation called *Plasmodium*

120. Primitive bacteria called methanogens

(1) Are photoautotrophic bacteria

(2) Have chitin in their cell wall

(3) Are abundant in marshy areas

(4) Show aerobic respiration

121. Lichen is association between

(1) Fungi and roots of higher plants

(2) Two autotrophic partners

(3) Algae and cyanobacteria

(4) Autotrophic and heterotrophic partners

122. Select the **correct** statement w.r.t. kingdom Protista.

(1) Boundaries of this kingdom are well defined

(2) All protists have cell wall

(3) Cell fusion and zygote formation does not occur

(4) Membrane bound organelles are present in protists

123. Read the below given assertion and reason and select the correct option.
Assertion (A) : Mixotrophic nutrition is seen in *Euglena*.
Reason (R) : *Euglena* is photosynthetic as well as chemosynthetic autotroph.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

124. Coprophilous fungi grow on

(1) Bark of trees

(2) Cow dung

(3) Burnt wood

(4) Dead animal bodies

125. Read the following statements and select the **correct** option.
Statement A : Bacteria as a group show most extensive metabolic diversity.
Statement B : Most of the bacteria are photosynthetic.

(1) Only A is correct

(2) Only B is correct

(3) Both A and B are correct

(4) Both A and B are incorrect

126. N_2 fixing cyanobacterium is

(1) *Pseudomonas*

(2) *Nostoc*

(3) *Rhizobium*

(4) *Chlorella*

127. Bracket fungi belongs to the class

(1) Phycomycetes

(2) Ascomycetes

(3) Basidiomycetes

(4) Deuteromycetes

128. Members of a class of fungi have branched and septate mycelium but no sex organs. They also produce

(1) Asexual planospores

(2) Sexual exogenous spores

(3) Sexual endogenous spores

(4) Asexual endogenous spores

129. Non-cellulosic cell wall is found

(1) Only in monerans

(2) Only in fungi

(3) Monerans and fungi

(4) Fungi and plants

130. The outer thin loose covering of some animal viruses composed of proteins, lipids and carbohydrates is called

(1) Envelope

(2) Capsid

(3) Capsomere

(4) Nucleoid

131. Read the following statements and select the option that is true for them.
Statement A: Amongst all eukaryotic chlorophyll-containing organism, a few members are partially heterotrophic.
Statement B: Viruses are characterised by having an inert crystalline structure outside the living cell.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements A and B are correct

(4) Both statements A and B are incorrect

132. Cyanobacteria differ from archaebacteria in having

(1) Nucleoid with nuclear membrane

(2) Nucleoid without nuclear membrane

(3) Chlorophyll *a* similar to green plants

(4) Cellulosic cell wall

133. In Zygomycetes, sporangiospores are

(1) Produced inside sporangia and are non motile

(2) Produced inside sporangia and are motile

(3) Nonmotile and exogenously produced

(4) Motile and exogenously produced

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134. State True **(T)** or False **(F)** to the given statements and select the **correct** option.
A. Lichens grow well in polluted area
B. Algal component of lichen is mycobiont and fungal component is phycobiont.

	A	B
(1)	T	T
(2)	F	T
(3)	T	F
(4)	F	F

(1) (1)

(2) (2)

(3) (3)

(4) (4)

135. The smallest prokaryotes that can survive without oxygen

(1) Are very complex in structure

(2) Are multicellular and filamentous

(3) Lack cell wall

(4) Cannot pass through bacterial filters

ZOOLOGY

136. Which of the following is a chronic disorder in which alveolar walls are damaged and due to which respiratory surface is decreased?

(1) Asthma

(2) Emphysema

(3) Silicosis

(4) Asbestosis

137. How many of the factors given in the box below favour(s) the dissociation of oxygen from the oxyhaemoglobin?

Low pCO_2, Low pH, High pO_2, Higher temperature

Select the **correct** option.

(1) Two

(2) One

(3) Three

(4) Four

138. In humans, there is a specialised centre called respiratory rhythm centre which is primarily responsible for the regulation of respiration. This respiratory rhythm centre is present in

(1) Medulla oblongata

(2) Aortic arch

(3) Cerebellum

(4) Pons

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139. Each adult human haemoglobin molecule contains _____ chains which can carry a maximum of _____ molecules of O_2.
Fill in the blanks respectively with the **correct** option.

(1) 2α and 1γ, two

(2) 2α and 2β, four

(3) 2α and 1β, two

(4) 1α and 2β, four

140. The event represented by the QRS-complex in a standard ECG of a normal adult man is the

(1) End of the ventricular systole

(2) Excitation of the atria

(3) Repolarisation of the ventricles

(4) Depolarisation of the ventricles

141.Assertion (A): In humans, the mature RBCs are enucleated and biconcave disc shaped.
Reason (R): Absence of nucleus provides more space for oxygen binding pigment haemoglobin and biconcave shape increases the total surface area for exchange of gases.
In the light of above statements, choose the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

142. The most abundant cells of the total WBCs and an agranulocyte that forms 6-8% of the total WBCs, both provide innate immunity.
Identify the cells and select the **correct** option w.r.t. them.

(1) Former has a kidney-shaped nucleus

(2) They both are phagocytic in nature

(3) Categorized under cytokine barriers of innate immunity

(4) Secrete histaminase and serotonin

143. In humans, under normal physiological conditions, cardiac output can be expressed as:

(1) Stroke volume + End systolic volume

(2) Stroke volume – End systolic volume

(3) Stroke volume × Heart rate

(4) Stroke volume × 2 (Heart rate)

144. In a standard ECG, the T-wave represents

(1) Depolarisation of atria

(2) Repolarisation of ventricles

(3) Depolarisation of ventricles

(4) Electrical excitation of atria

145. Carbon dioxide combines with haemoglobin in order to form

(1) Carboxyhaemoglobin

(2) Oxyhaemoglobin

(3) Myoglobin

(4) Carbaminohaemoglobin

146. Under normal physiological conditions in human beings, every 100 mL of oxygenated blood can deliver _____ mL of O_2 to the tissues. Select the correct option to fill in the blank.

(1) 4 mL

(2) 10 mL

(3) 2 mL

(4) 5 mL

147. Primary site of exchange of gases in humans is

(1) Bronchi

(2) Alveoli

(3) Bronchiole

(4) Trachea

148. In athletes, all of the following observations are seen compared to a normal person during resting conditions, **except**

(1) Decrease in heart rate

(2) Heart muscles become stronger through regular exercise

(3) Increase in stroke volume

(4) Decrease in cardiac output

149. Binding of oxygen with haemoglobin is primarily related to which of the following factors?

(1) Partial pressure of CO_2

(2) Partial pressure of O_2

(3) Hydrogen ion concentration

(4) Temperature

150. The partial pressure of O_2 and CO_2 in the systemic veins is same as that in

(1) Systemic arteries

(2) Pulmonary arteries

(3) Alveolar air

(4) Atmospheric air

151. The solubility of CO_2 is _____ times higher than that of O_2 in blood plasma of humans. Fill in the blank with **correct** option.

(1) 10-15

(2) 50-55

(3) 30-35

(4) 20-25

152. Consider a case where the cardiac output of a person is 5250 mL and the stroke volume is 70 mL. What would be the duration of the cardiac cycle in this person?

(1) 0.90 second

(2) 0.80 second

(3) 0.45 second

(4) 1.20 seconds

153. Which of the following conditions decreases the cardiac output in humans?

(1) Release of epinephrine from adrenal medulla

(2) Neural signals through the sympathetic nerve fibres

(3) Neural signals through the parasympathetic nerve fibres

(4) Release of noradrenaline from the adrenal medulla

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154. Plasma is well defined as

(1) Blood without formed elements

(2) Formed elements without clotting factors

(3) Blood with clotting factors

(4) Blood without platelets

155. In hepatic portal system, the hepatic portal vein carries blood from

(1) Kidney to the liver

(2) Intestine to the liver

(3) Liver to heart

(4) Liver to intestine

156. Stroke volume can be best described as

(1) Volume of blood that remains in each ventricle after ventricular systole.

(2) Volume of blood pumped out by each ventricle of heart per minute

(3) Cardiac output × Heart rate

(4) Volume of blood pumped out by each ventricle during a cardiac cycle

157. In blood vessels, the tunica intima consists of

(1) Smooth muscle and elastic fibres

(2) Fibrous connective tissue

(3) Squamous endothelium

(4) Smooth muscle and fibrous connective tissue

158. Select the **correct** match.

	Leucocytes		**Percentage in the total leucocytes**
(1)	Monocytes	–	0-1%
(2)	Eosinophils	–	2-3%
(3)	Basophils	–	2-10%
(4)	Neutrophils	–	20-25%

(1) (1)

(2) (2)

(3) (3)

(4) (4)

159. How many of the given statements is/are correct w.r.t. platelets in humans?
(A) They are nucleated cells produced from megakaryocytes.
(B) Blood contains 1,500,00–3,500,00 platelets mm^{-3}.
(C) Their reduction leads to excessive blood loss from the body on injury.
(D) They constitute formed elements along with erythrocytes and leucocytes.
Select the **correct** option.

(1) One

(2) Two

(3) Three

(4) Four

160. Complete the analogy w.r.t. ABO blood grouping in humans and select the correct option.
Universal donor : Blood group _____ : : Universal recipient : Blood group AB.

(1) 'O'

(2) 'B'

(3) 'A'

(4) 'AB'

161. In humans, the respiratory disorder named asthma may attributed to

(1) Reduced respiratory surface of alveoli

(2) Lung fibrosis

(3) Exaggerated inflammation of bronchioles

(4) Increased surface area of lungs

162. Consider the following features:
a. Made up of three layers
b. Total thickness is more than a millimetre
c. One of the layer is made up of thin cuboidal epithelium
Select the option that represents the **incorrect** feature(s) w.r.t. the diffusion membrane in human lungs.

(1) a, b and c

(2) a and b

(3) Only a

(4) b and c

163. Partial pressure of a gas

(1) Is the pressure contributed by an individual gas in a mixture of gases

(2) Is the pressure contributed by all gases in a mixture of gases

(3) Cannot affect the rate of diffusion of gases

(4) Present in atmosphere is equal to the atmospheric pressure

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164. In humans, CO_2 from blood crosses how many acellular layer(s) of diffusion membrane to reach the lumen of alveoli?

(1) Two

(2) Three

(3) One

(4) Four

165. Select the **correct** option to complete the analogy w.r.t. humans
Transport of O_2 by RBCs in blood : 97% : : Transport of CO_2 by RBCs in blood : ____

(1) 70%

(2) 3%

(3) 20-25%

(4) 7%

166. Enzyme that facilitates the formation of carbonic acid at tissue level and is present in high concentration within RBCs, is

(1) Carbonic anhydrase

(2) Carboxypeptidase

(3) Peroxidase

(4) Catalase

167. Minimum amount of O_2 and CO_2 is transported by/as

(1) Bicarbonates

(2) RBCs

(3) Being dissolved in plasma

(4) Combining with haemoglobin

168. Choose the correct statement w.r.t the cardiac cycle of a healthy individual under normal physiological conditions.

(1) The volume of blood pumped out by the right and left ventricles is same.

(2) The volume of blood pumped out by the right and left ventricles is significantly different.

(3) Complete filling of ventricles occur during joint diastole.

(4) Amount of blood pumped out in each cardiac cycle is called cardiac output.

169. The type of portal system **absent** in humans is
a. Renal
b. Hepatic
c. Hypophyseal
Select the **correct** option.

(1) a & b

(2) b & c

(3) only a

(4) only c

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170. Choose the **odd** one with respect to components of blood.

(1) Fluid matrix

(2) Plasma

(3) Formed elements

(4) Fibres

171. The life span of the erythrocytes in humans is about

(1) 120 days

(2) 150 days

(3) 190 days

(4) 180 days

172. Deoxygenated blood from the wall of heart is provided to right atrium by

(1) Coronary sinus

(2) Inferior vena cava

(3) Superior vena cava

(4) Pulmonary artery

173. Under normal conditions, atrial systole

(1) Increases the flow of blood into the ventricles by 70 percent

(2) Is due to generation of action potential in AVN

(3) Increases the flow of blood into the ventricles by 30 percent

(4) Coincides with ventricular systole

174. Select the **incorrect** statement w.r.t. human heart.

(1) During joint diastole, blood from the pulmonary veins and vena cava flows into the right and left ventricle respectively.

(2) Increase in ventricular pressure leads to opening of semilunar valves guarding the pulmonary artery and aorta.

(3) As the ventricular pressure declines, the AV valves are pushed open by the pressure in the atria exerted by the blood which is being emptied into them.

(4) Atrial diastole coincides with the ventricular systole.

175. Bicuspid valve is present between

(1) Right and left atrium

(2) Right and left ventricle

(3) Right atrium and right ventricle

(4) Left atrium and left ventricle

176. Which of the following is correct w.r.t. the second heart sound *i.e.*, dub during a cardiac cycle?

(1) It it associated with the closure of semilunar valves at the begining of ventricular systole

(2) It is produced after an interval of 0.8 seconds from the first heart sound

(3) It is lower pitched and cannot be easily heard through a stethescope

(4) It is associated with the closure of semilunar valves at the begining of ventricular diastole

177. Which of the following statements is not true about blood pressure?

(1) Blood pressure is measured with an instrument called sphygmomanometer

(2) If repeated checks of the blood pressure of an individual result in 140/90 mm Hg or higher, it shows hypertension

(3) The normal systolic pressure is 120 mm Hg and diastolic pressure is 80 mm Hg.

(4) Hypertension is caused by vasodilation which results in increased resistance to blood flow

178. Which of the following representations is **false** w.r.t. partial pressure of O_2 and CO_2 under normal physiological conditions in human?

(1) In alveoli, $pCO_2 < pO_2$

(2) In deoxygenated blood, $pCO_2 > pO_2$

(3) In tissues, $pCO_2 > pO_2$

(4) In atmospheric air, $pO_2 < pCO_2$

179. Two separate blood circulatory pathways are present in

(1) Birds, mammals

(2) Earthworms, flatworms

(3) Sponges, fishes

(4) Amphibians, reptiles

180. Assertion (A): Neural signals from pneumotaxic centre can reduce the duration of inspiration in human.
Reason (R): Pneumotaxic centre is located in pons region of hindbrain.
In the light of above given statements, choose the **correct** option from the following.

(1) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(2) Both (A) and (R) are true, (R) is the correct explanation of (A).

(3) (A) is true, (R) is false

(4) (A) and (R) are false

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